Exercise 2 - Aggregate Functions & Grouping

1. SELECT COUNT (id) AS total_number_of_employees
FROM employees;

total_number_of_employees
10

2. SELECT SUM (salary) AS total_salary,
department
FROM employees
WHERE department = 'IT';

total_salary	department
220 000	IT

3. SELECT department,
AVG(salary) AS average_salary
FROM employees
WHERE department = 'HR';

average_salary	department
49 500	HR

4. SELECT MAX (salary) AS highest_salary
MIN (salary) AS lowest_salary
FROM employees

highest_salary	lowest_salary
62 000	48 000

5. SELECT department,
SUM (salary) AS total_salary
FROM employees
GROUP BY department;

department	total salary
IT	220 000
HR	99 000
Finance	119 000
Marketing	105 000

6. SELECT city,
COUNT (id) AS total_employees
FROM employees
GROUP BY city;

total_employees	city
2	New York
3	Chicago
2	Los Angeles
2	San Francisco
1	Houston

7. SELECT department
AVG(salary
FROM employees
GROUP BY dep
ORDER average

8. SELECT depa
SUM~~AVG~~ (s
FROM emplo
GROUP BY
HAVING S

9. SELECT
C
FROM
~~WHERE~~
HAV
ORD

10

7. SELECT department,
AVG(salary) AS average_salary
FROM employees
GROUP BY department
ORDER average-salary BY DESC;

Average_salary	departments
59 500	IT
55 000	HR
52 500	Finance
49 500	Marketing

8. SELECT department,
SUM~~AVG~~(salary) AS ~~average~~ total_salary
FROM employees
GROUP BY department
HAVING SUM(salary) > 100 000

total_salary	departments
105 000	Marketing
119 000	Finance
220 000	IT

9. SELECT city,
COUNT(id) AS number_of_employees
FROM employees
~~WHERE~~ GROUP BY COUNT(id)
HAVING ~~COUNT(id)~~ ~~city~~ > 2
ORDER BY DESC

number_of_employee	city
2	New York
3	Chicago
2	Los Angeles
2	San Francisco

10. SELECT department,
MAX (salary) AS highest_salary
AVG (salary) AS highest_average_salary
FROM employees
GROUP BY department

department	highest_avg_salary
IT	59 500